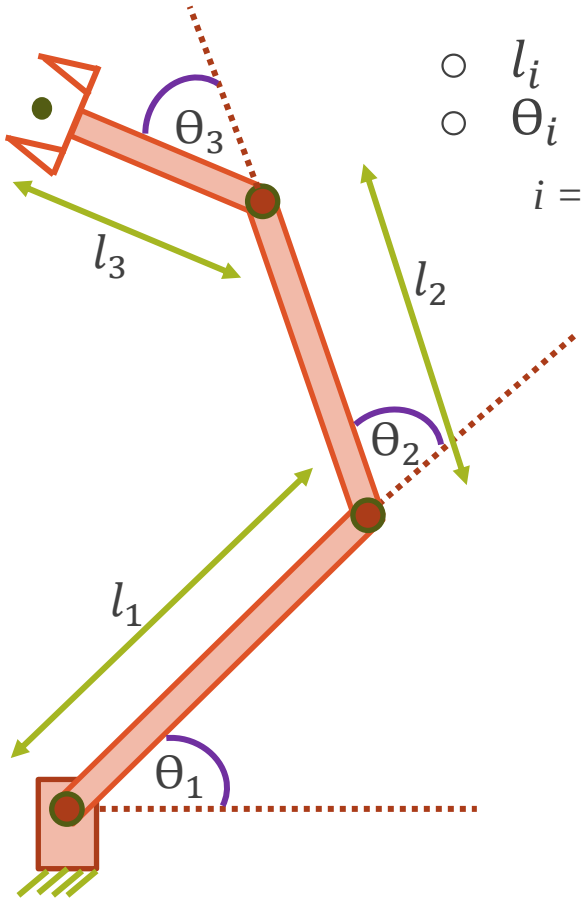
3R or RRR Planar Robotic Arm

Presented by: Sourav Kumar

3R or RRR Robotic Arm

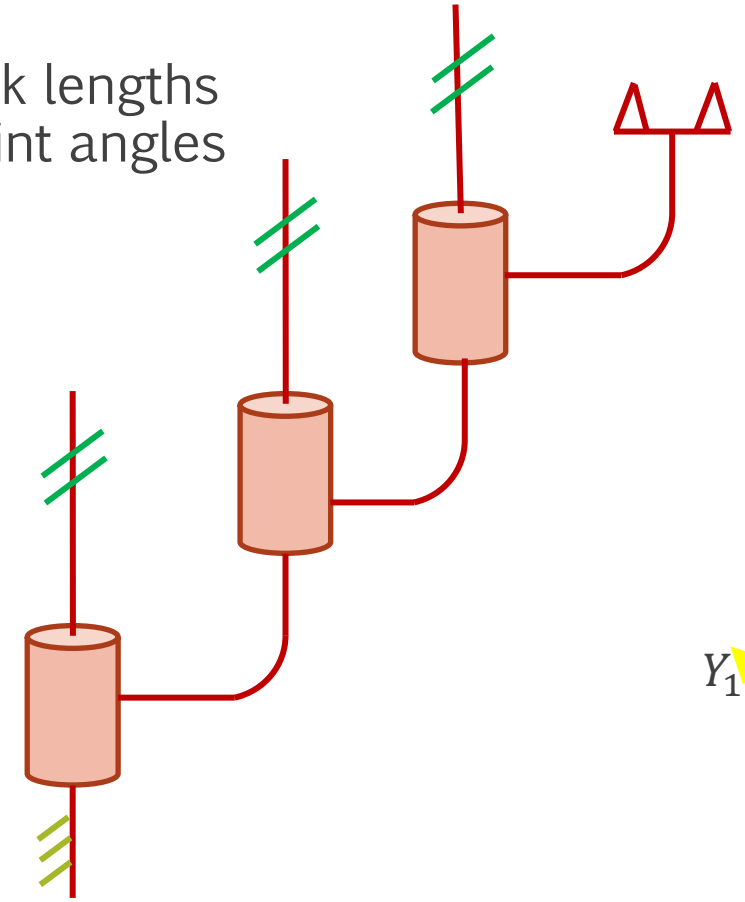
- All three joints are revolute.
  - Double hash marks indicates all 3 axes are parallel.



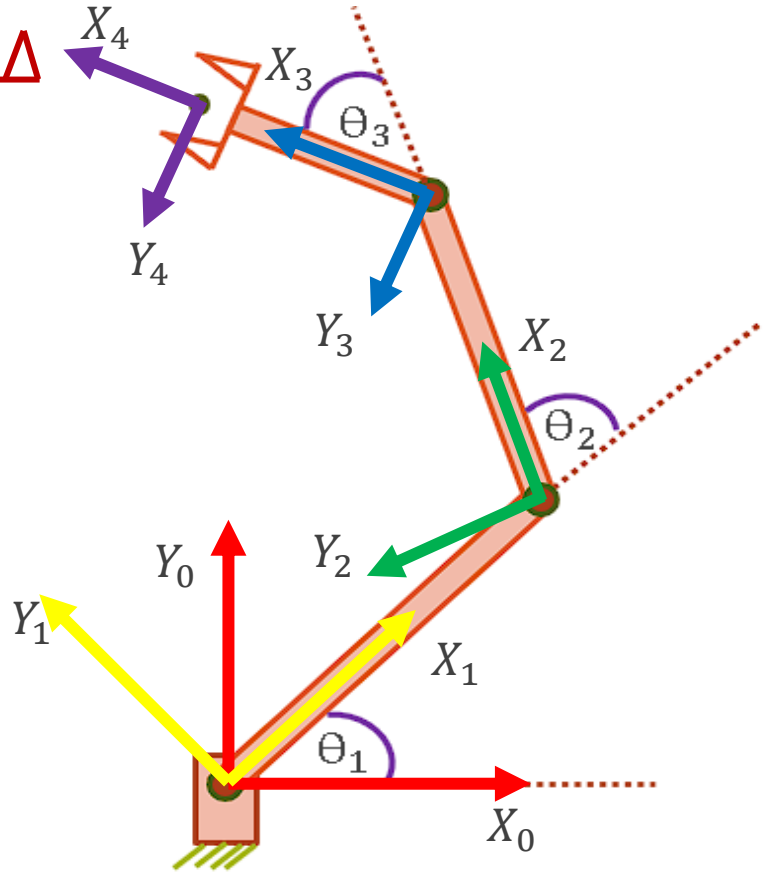
- l_i → Link lengths
- θ_i → Joint angles

$i = 1, 2, 3$

Kinematic diagram



Schematic Diagram

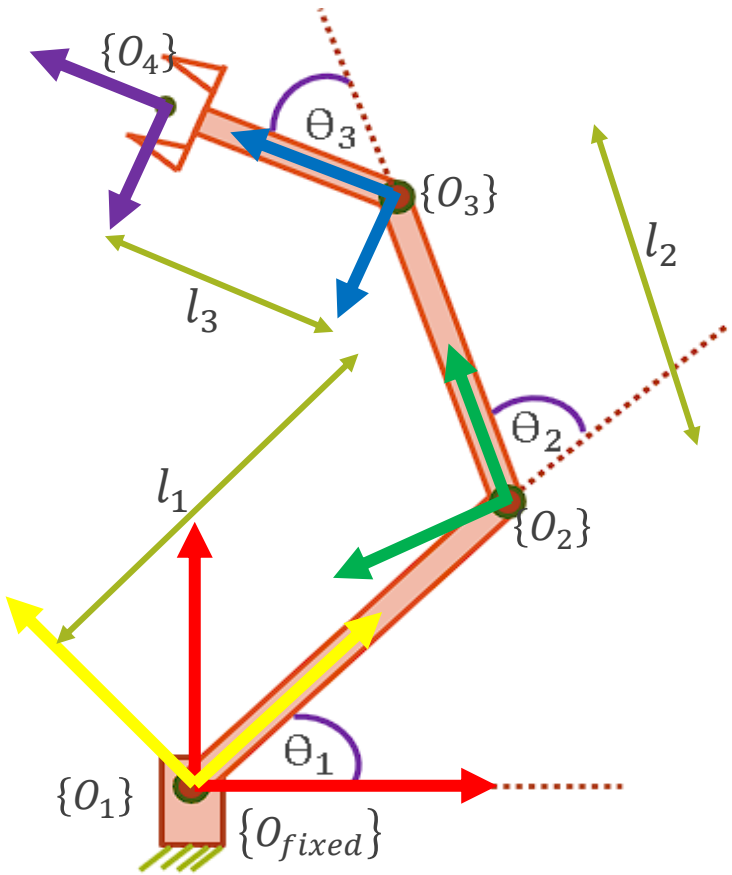


Assigning Frames

DH Parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	l_1	0	θ_2
3	0	l_2	0	θ_3
4	0	l_3	0	0

- $\{O_{fixed}\}\rightarrow$ Reference frame
- $\{O_1\}, \{O_2\}, \{O_3\} \rightarrow$ Intermediate frames
- $\{O_4\}\rightarrow$ Tip or End-Effector frame
- $\alpha_{i-1}\rightarrow$ Link twists
- $a_{i-1}\rightarrow$ Link lengths
- $d_i \rightarrow$ Link offsets
- $\theta_i \rightarrow$ Joint angles



Link Frame Assignment

Transformation of frames

- ${}^{i-1}_{i}[T]$ = Screw$_X(\boldsymbol{a_{i-1}}, \alpha_{\boldsymbol{i-1}})$ Screw$_Z(\boldsymbol{d_i}, \theta_{\boldsymbol{i}})$

$$= \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1} \\ \sin\theta_i \cos\alpha_{i-1} & \cos\theta_i \cos\alpha_{i-1} & -\sin\alpha_{i-1} & -\sin\alpha_{i-1} d_i \\ \sin\theta_i \sin\alpha_{i-1} & \cos\theta_i \sin\alpha_{i-1} & \cos\alpha_{i-1} & \cos\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ${}^{0}_{4}[T]$ = ${}^{0}_{1}[T]$ ${}^{1}_{2}[T]$ ${}^{2}_{3}[T]$ ${}^{3}_{4}[T]$

$$= \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & L_1 \\ \sin\theta_2 & \cos\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & L_2 \\ \sin\theta_3 & \cos\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \left[\begin{array}{c:c} {}^{0}_{4}[R]_{3X3} & \overrightarrow{{}^{0}_{4}p}_{3X1} \\ \hdashline 0_{3X1} & 1 \end{array}\right]_{4 X 4}$$

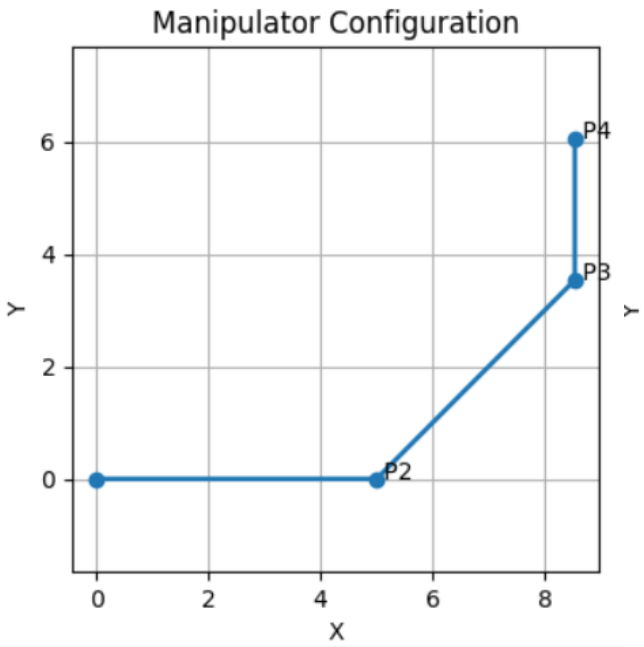
- ${}^{0}_{4}[R]_{3X3}$, represents orientation of tip w.r.t. fixed base
- $\overrightarrow{{}^{0}_{4}p}_{3X1}$, represents the position of tip/end effector w.r.t. fixed base

Results

Theta 1 (degrees)
0
Theta 2 (degrees)
45
Theta 3 (degrees)
45
Compute & Plot

P1 = [0. 0. 0.]
P2 = [5. 0. 0.]
P3 = [8.53553391 3.53553391 0.]
P4 (Tip) = [8.53553391 6.03553391 0.]

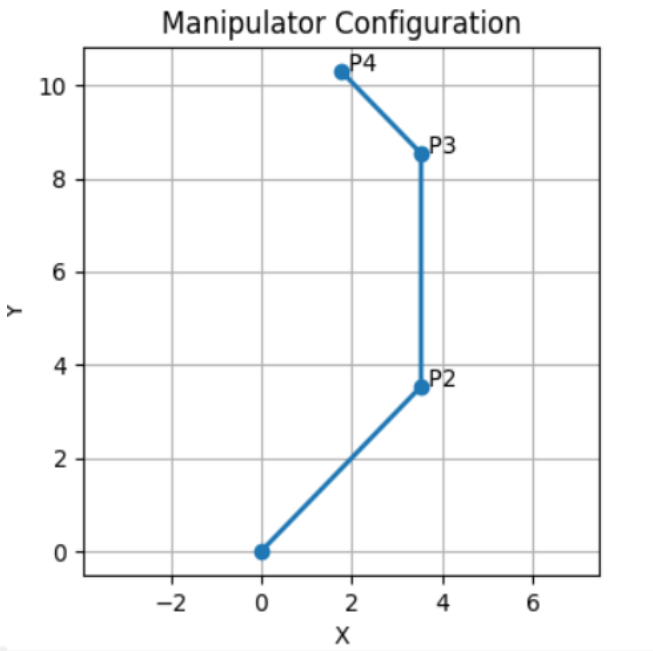
Final Orientation (deg) = 90.00



Theta 1 (degrees)
45
Theta 2 (degrees)
45
Theta 3 (degrees)
45
Compute & Plot

P1 = [0. 0. 0.]
P2 = [3.53553391 3.53553391 0.]
P3 = [3.53553391 8.53553391 0.]
P4 (Tip) = [1.76776695 10.30330086 0.]

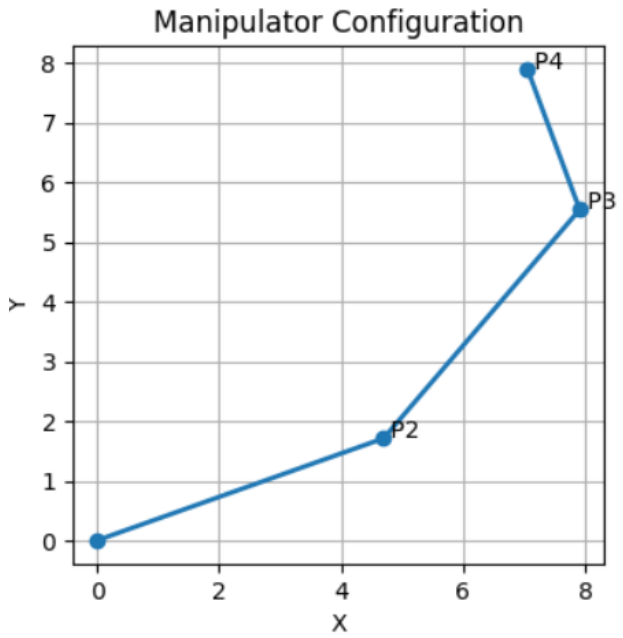
Final Orientation (deg) = 135.00



Theta 1 (degrees)
20
Theta 2 (degrees)
30
Theta 3 (degrees)
60
Compute & Plot

P1 = [0. 0. 0.]
P2 = [4.6984631 1.71010072 0.]
P3 = [7.91240115 5.54032293 0.]
P4 (Tip) = [7.05735079 7.88955448 0.]

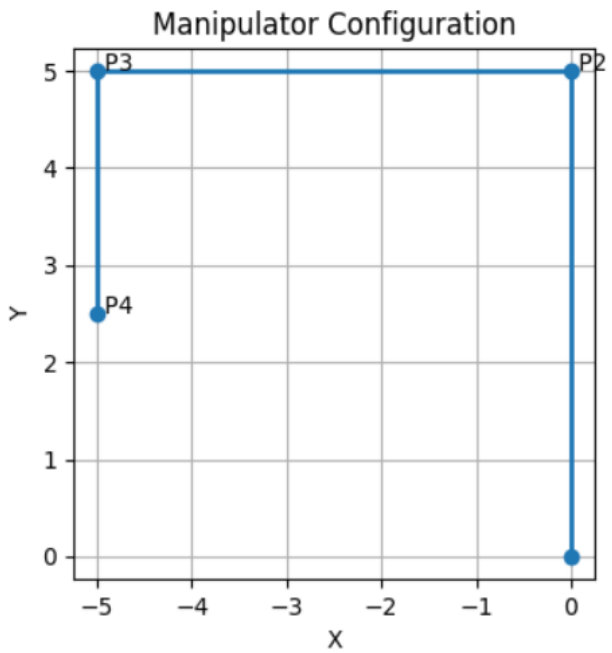
Final Orientation (deg) = 110.00



Theta 1 (degrees)
90
Theta 2 (degrees)
90
Theta 3 (degrees)
90
Compute & Plot

P1 = [0. 0. 0.]
P2 = [3.061617e-16 5.000000e+00 0.000000e+00]
P3 = [-5. 5. 0.]
P4 (Tip) = [-5. 2.5 0.]

Final Orientation (deg) = -90.00



Thank You